

ARCHEÍO

File Utility Platform

README — Milestone 1 Submission

Proposed Level of Achievement

Apollo 11

Motivation

Modern workflows frequently involve scanned documents, screenshots, lecture slides, receipts, and PDFs that contain text which cannot be searched or edited. Users often need to manually retype information or rely on multiple fragmented online tools to convert files, compress documents, or extract text.

Existing solutions are typically scattered across different websites, many of which contain intrusive advertisements, limited functionality, or raise privacy concerns when users upload sensitive documents.

This project aims to build a **unified, privacy-first file utility platform** that allows users to unlock trapped data in images and PDFs. Through the integration of OCR, format conversion, and intelligent compression, users can convert static documents into searchable and editable formats.

By providing a **single streamlined interface for multiple document utilities**, the system reduces friction, improves productivity, and eliminates the need for users to rely on multiple third-party tools.

Aim

We aim to develop a **web and mobile application that consolidates common document utilities into a single integrated system**. The platform will enable users to convert images and PDFs into searchable documents, transform files between formats, and compress files efficiently.

The goal is to make document processing **faster, simpler, and more secure**, while demonstrating how multiple backend processing services can be integrated into a cohesive application.

User Stories

Student

As a **student** who receives documents as scanned PDFs, I want to **convert the document into searchable text**, so that I can **quickly locate useful information using Ctrl/Cmd + F while studying**.

Office Worker

As an **office worker handling scanned documents**, I want to **convert images or PDFs into editable Word documents**, so that I can **modify and reuse the content easily without retyping everything**.

Researcher

As a **researcher reviewing large numbers of PDFs**, I want to **make scanned papers searchable**, so that I can **quickly locate important sections across documents**.

General User

As a **general user managing many files**, I want to **compress images and documents**, so that I can **reduce file size before uploading or sharing them online**.

Milestone Plans

The following outlines the planned features and deliverables for each milestone of the project.

Milestone 1 — Proof of Concept [Completed]

The goal of Milestone 1 is to establish a working proof of concept, with a functional website UI and support for basic file processing across common media types.

- Developed a functional draft of the website user interface (UI)
- File format conversion for basic media types:
 - Images — formats supported by the HTML5 Canvas API in Vite (e.g. PNG, JPEG, WebP, BMP, GIF)
 - Videos — basic video format conversion
 - Audio — basic audio format conversion
- File compression for images, videos, and audio

Milestone 2 — Extended Support & Basic Editing

Milestone 2 expands on the proof of concept by adding support for more file types and introducing basic media editing features.

- Expanded file format support beyond canvas-compatible formats (e.g. additional image codecs, more video and audio containers)
- Basic image editing:
 - Cropping, rotating, and resizing
 - Brightness and contrast adjustments
- Basic video editing:
 - Trimming and cutting clips
 - Resizing and resolution adjustment

- Basic audio editing:
 - Trimming and cutting
 - Volume and playback speed adjustment

Milestone 3 — Advanced Document Processing

Milestone 3 introduces advanced document handling capabilities, with a focus on PDF manipulation and OCR-based text extraction.

- PDF editing:
 - Rearranging, merging, and splitting pages
 - Basic annotation support
- OCR integration:
 - Convert scanned PDFs into fully searchable documents, enabling Ctrl/Cmd + F
 - Extract text layers and generate editable Word documents

Poster Submissions

Liftoff

Our poster submission during Liftoff presents the basic scaffolding of our proposed features after extensive brainstorming and designing. We were still experimenting with various colour schemes and user interface ideas.

Mission Control 1

We now have a much clearer vision of the product we want to build, and we believe that the white and beige colour scheme effectively conveys the concept of working with digital tools in a way that feels familiar and tangible, while maintaining a clean and modern user interface. We also discovered that certain file conversions, such as DOCX to PDF, cannot be performed entirely on the user's device. As a result, we updated our poster to clarify that most file conversions will be processed locally rather than all conversions.

Video Submissions

Milestone	Video Link
Liftoff	https://tinyurl.com/archeioliftoff
Milestone 1	https://tinyurl.com/archeioMS1
Milestone 2	—
Milestone 3	—

Liftoff

We created a promotional video to introduce our file conversion app. We showcased our motivation and aim, and demonstrated an array of features we hope to implement that will allow users to seamlessly convert, compress, manipulate, and unlock files.

Milestone 1

As we were focused on developing other core features of the app and continuously refining the user interface, we decided to reuse the same video from Liftoff for our Milestone 1 submission. Since the changes made thus far were mainly minor UI refinements and updates to the colour scheme, we chose to prioritise further improvements to the app's functionality before producing an updated walkthrough video.

Milestone 2

To be updated.

Milestone 3

To be updated.

Features

Core Features

1. Document and Media Format Conversion

Convert between common document and media formats.

- Example: Image → Word → PDF

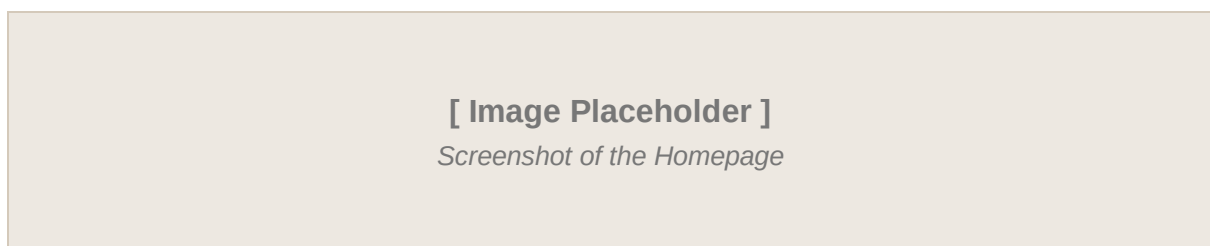


Figure 1: Draft of the Archeio homepage

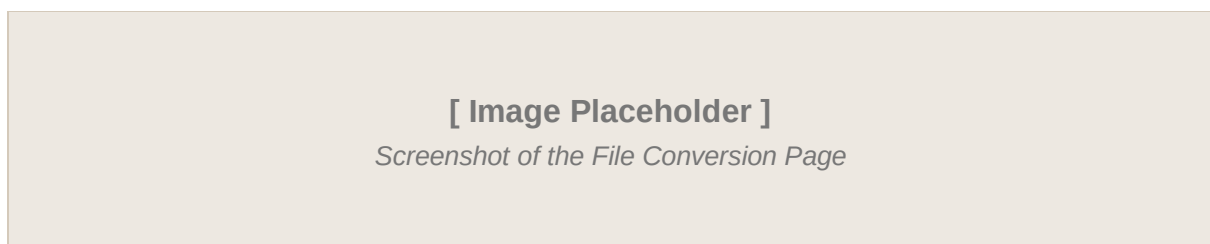


Figure 2: Draft of the file conversion page

2. File Compression

- Reduce image file size by adjusting compression level or readjusting aspect ratios
- Compress files into archive formats such as ZIP
- Media resolution and size reduction



Figure 3: Draft of the file compression page

3. File Editing

- Basic editing of images (cropping and scaling)
- Basic editing of PDFs (rearranging pages and combining PDFs)

4. Batch Processing

- Process multiple files simultaneously

5. Smart Searchable PDFs

- Automatically overlay OCR text onto scanned PDFs to enable search functionality

6. Privacy-first Processing

- Temporary file storage with automatic deletion after processing
- Locally run file processing — sensitive files never leave the user's device

7. OCR-based Document Unlocking

- Convert images or scanned PDFs into complete, searchable text documents
- Extract text and generate editable Word documents

8. Audio to MIDI Converter

- Convert audio files (e.g. MP3) to MIDI format

Tech Stack

Frontend

- React.js and React Native
- HTML / CSS
- Tailwind CSS

Backend

- Python (FastAPI / Flask)

Document Processing

- Tesseract OCR
- Python libraries for PDF manipulation (e.g. PyPDF / pdfplumber)
- LLM integration

File Processing

- Pillow / OpenCV for image processing
- ZIP utilities for file compression

Infrastructure

- Docker (containerization)
- Cloud deployment (e.g. AWS / Vercel)